Students's Name: ……

Signature: ……Class & Stream: ……..……..

P510/1
PHYSICS
Paper 1
(Theory)
2 hours 15 minutes

UGANDA STEM TEACHERS NETWORK (USTEN)

Uganda Advanced Certificate of Education

PHYSICS REVISION ASSESSMENT SET THREE

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- This examination paper consists of **THREE compulsory** items.
- Responses to **each** item should start on a **fresh page**.
- Silent non-programmable scientific calculators may be used.

ITEM 1

A regional hospital plans to expand its nuclear medicine department to provide treatment for patients with thyroid disorders. The department intends to use radioactive iodine-131, $^{131}_{53}I$, which undergoes beta decay and also emits gamma radiation.

Before the new facility becomes operational, the hospital's radiation-protection team wants to determine whether the proposed quantity of iodine-131 satisfies the specified activity and energy requirements.

The hospital also plans to acquire an X-ray imaging system for examining patients before and after treatment. An Advanced Level Physics learner has been requested to explain the physical principles involved in the production of X-rays.

Support material

- Mass of $^{131}_{53}I$ = 130.90612 U
- Mass of $^{131}_{54}Xe$ = 130.90508 U
- Half-life of $^{131}_{53}I$ = 8.02 days
- Mass of proposed sample = 5.0 g
- Molar mass of ^{131}I = 131 g mol^{-1}
- Avogadro constant = 6.02×10^{23} mol^{-1}
- 1 U = 931 MeV/c^2
- Maximum permitted activity = 6.5×10^{13} Bq
- Minimum useful energy per decay = 0.90 MeV

Task

You have been selected to assist the hospital's radiation-protection team.

a) Analyse the supplied data to determine whether the proposed iodine-131 sample meets the specified requirements.
b) The hospital intends to use an X-ray imaging system for diagnostic examinations. Explain the construction and working principle of an X-ray tube, describing how accelerated electrons produce X-rays and how the radiation can be used to form an image.

ITEM 2

A national environmental monitoring programme is preparing to launch a weather satellite to provide information about rainfall, cloud cover and severe weather systems. The satellite has a mass of 250 kg and is required to enter a circular orbit in which its total mechanical energy is -5.0×10^9 J.

The mission engineers need to determine the height of the parking orbit and assess how the satellite's mechanical energy is related to its orbital radius.

Before the launch, a maintenance technician must travel from the engineering workshop to the tracking station to deliver calibration equipment. He uses a vehicle of mass 1600 kg. The vehicle engine supplies a driving force of 900 N and has a power output of 16.2 kW.

The vehicle begins from rest and accelerates uniformly until it reaches the speed corresponding to the engine's power and driving force. The acceleration stage covers 0.75 km. The technician then applies the brakes over the remaining 0.60 km.

The coefficient of friction between the tyres and the road is 0.035. The technician has 5 minutes to complete the journey.
The environmental team also wants to obtain aerial images of a forest reserve. The supervisor has asked the technician to explain how a lightweight locally assembled flying device carrying a camera could generate enough lift to remain in the air.

Support Material

- Mass of satellite = 250 kg
- Required mechanical energy = -5.0×10^9 J
- Radius of Earth, RE = 6.4×10^6 m
- g = 9.81 m s^{-2}
- Vehicle mass = 1600 kg
- Driving force = 900 N
- Engine power = 16.2 kW
- Coefficient of friction = 0.035
- Acceleration distance = 0.75 km
- Braking distance = 0.60 km
- Available time = 5 minutes

Task

You have been selected to assist the environmental monitoring team.

a) Analyse the data provided and determine whether the proposed arrangements are adequate.
b) Explain how a camera-carrying quadcopter can be designed to produce lift and describe the physical principles responsible for its flight and control.

ITEM 3

A hotel in Kampala recently installed a large double-glazed glass frontage in its conference room. The management expected the glazing to reduce heat entering the room while allowing sufficient daylight.
However, guests began complaining that the conference room became very hot during periods of strong sunshine. The supplier maintained that the glazing met the required standards and that when the outside surface of the glass reached 36°C, the temperature inside the room would not exceed 23°C.
The glazed frontage consists of two glass panes, each 4 mm thick, separated by a 12 mm layer of air. The total area of the glazing is 20 m^2. During a sunny period, the glazing receives solar radiation of intensity 1000 W m^{-2}, and 50% of the incident radiation is absorbed.
The supplier quoted the thermal conductivity of the glass as 0.90 W m^{-1} K^{-1}, while the thermal conductivity of the air layer was 0.025 W m^{-1} K^{-1}.
To verify the manufacturer's claim, a Senior Six Physics learner investigated a sample of the glass. The sample had a cross-sectional area of 0.060 m^2 and thickness 15 mm. When a steady temperature difference of 10 K was maintained across the sample, the rate of heat transfer was measured to be 36 W.
The hotel plans to modify the outside surface of the glass to reduce solar heating. It also wishes to improve security by installing a device near the conference-room entrance capable of detecting movement.

Support Material

- Thermal conductivity of air = 0.025 W m^{-1} K^{-1}
- Quoted thermal conductivity of glass = 0.90 W m^{-1} K^{-1}
- Solar radiation intensity = 1000 W m^{-2}
- Fraction of radiation absorbed = 50%
- Glazing area = 20 m^2

Task

You have been selected by the Physics teacher.

a) Using the data provided, analyse the situation and establish whether the supplier's claims concerning the thermal insulation of the glazing are supported by the Physics. In your analysis, determine the thermal conductivity of the glass sample and use the relevant thermal-conduction principles to assess the window.

b) Explain a suitable modification of the external surface of the glass that could reduce the absorption of solar energy and describe the operation of an infrared motion-sensing security device that could be installed near the entrance.

END

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